

Product Requirements Document (PRD)

Product: Unknown — algebra learn-by-doing app

Version: 1.0 — Phase 1 MVP · **Updated:** June 22, 2026

Assignment: Build Brilliant — A Learn-by-Doing App for One Subject

Phase scope: Phase 1 only. No AI. No Phase 2/3 features.

1. Problem statement

Most learning apps show Maya a video, then a quiz. She memorizes steps, gets a red X when wrong, and still does not understand **why** `x + 3 = 7` works. In class she is afraid to ask questions. At home she needs something that lets her **try, fail safely, and figure it out herself** — in five to ten minutes, on whatever device is nearby.

This product **is a Brilliant clone in the ways that matter for learning** — but only the key aspects, scoped to one persona. It is not feature-parity with the full Brilliant platform.

2. Product vision

A responsive web app where Maya signs in, works through a short linear course of interactive algebra lessons, manipulates balance scales and tiles, gets instant kind feedback when she is wrong, passes a no-hint mastery check, and sees her streak grow — on her phone after school or on a laptop at the kitchen table. The app teaches without AI and proves the core experience works before anything smarter is added.

2.1 Relationship to Brilliant

The assignment asks us to model the product on Brilliant. **Technically, this is a Brilliant clone — but deliberately limited to the core mechanics that make Brilliant work for a learner like Maya**, not a copy of the entire platform.

Brilliant aspect we clone	How we implement it
Learn by doing	Interactive problems, not video lectures
Action before theory	Shapes on scales before introducing <code>x</code>
Balance-scale algebra	Primary visual metaphor (Solving Equations course)
Scaffolded → mastery rhythm	Hand-holding questions + explanation pages, then 3 no-hint mastery Q
Instant, specific feedback	Pre-written copy on every answer
Visual / manipulable lessons	Scale drag/tap, tile combine (≥1 interactive per lesson)
Structured content model	Lessons as JSON steps, not HTML blobs
Course path	Linear unlock through 5–6 lessons in one subject
Habit loop	Daily streak, continue CTA, lesson-complete moment

Brilliant aspect we do not clone	Why
Full course library (50+ lessons, 1000+ problems)	Depth for Maya, not breadth
Koji AI tutor	Phase 2; banned in MVP
Leagues, XP, levels, badges	Streak + completion is enough for her
Multiple subjects and algebra "lenses"	One subject, one path
Practice sets, review sets, spaced repetition	Phase 3
Premium tiers, educator dashboard, offline	Wrong scope for MVP

Positioning: A focused Brilliant-style experience for early algebra — built for Maya, not a platform rebuild.

3. Domain

3.1 Subject

Algebra — early solving equations and expressions, using a balance-scale and tile mental model.

3.2 Course

Title: Solving Equations (Intro) · **Audience:** Maya only. All copy, difficulty, and tone tuned for her.

3.3 In scope (mathematical)

Topic	Examples
Finding unknowns (concrete)	Shapes on scales before x
Variables	x as a name for an unknown
One-step equations	$x + 3 = 7$, $x - 2 = 6$, $10 = x + 4$
Combining like terms	$2x + x = 3x$, then solve
Writing expressions	"5 more than a number" $\rightarrow x + 5$
Two unknowns (optional L6)	$x + y = 10$, $x = 4$ — no coefficients

Constraints: whole numbers only; coefficients ≤ 3 ; no negatives, fractions, distributive property, graphing, or two-step equations.

3.4 Out of scope (mathematical)

Two-step equations, inequalities, distributive property, factoring, quadratics, graphing, word-problem systems, anything requiring a calculator.

4. User persona: Maya

Attribute	Detail
Age / grade	12–13, 7th grade
Context	First algebra unit; has seen x but does not trust it
Mindset	Anxious about being wrong; avoids asking for help
Learning style	Needs to *do* something before a rule makes sense
Session length	5–10 minutes
Devices	Phone after school; laptop/desktop at home
Goal	Understand what x means; solve basic one-step equations
Success signal	Wrong → hint → fix herself → finish lesson → return tomorrow

Design principles

- 1. **Concrete before abstract** — shapes on scales before the symbol x .
- 2. **Warm, never condescending** — no "obvious!", no jargon-first explanations.
- 3. **Short and completable** — each lesson fits one sitting; clear "you're done!" moment.
- 4. **Safe to fail** — mastery allows 2/3 pass so one mistake does not destroy confidence.
- 5. **Predictable rhythm** — every lesson: scaffolded practice, then mastery check.
- 6. **Device-agnostic** — phone and computer; layout adapts; touch and mouse both work.

Not designing for

Exam crammers, teachers managing classrooms, multi-subject learners, users who want video lectures or AI chat tutors.

5. MVP definition

5.1 One-sentence MVP

Maya can sign in on **mobile or desktop**, complete **5 interactive algebra lessons** (each with scaffolded practice and a 3-question mastery check), manipulate scales/tiles, get instant written feedback, resume mid-lesson, see her streak and next lesson — deployed publicly with **zero AI**.

5.2 Assignment alignment (Phase 1 gate)

Requirement	Implementation
Chosen subject + persona	Algebra + Maya (stated in README)
Interactive lesson(s)	5 lessons (depth in one subject)
Direct manipulation	≥ 1 interactive step per lesson (scale/tiles)
Visual responds in real time	Scale/tiles update on interaction
Instant, specific feedback	Pre-written copy; UI response $< 100\text{ms}$
Progress persists	Firestore; resume mid-lesson and mid-phase
Auth + names	Firebase Auth + display name
Mobile + desktop	Responsive web app (§9)
Deployed & public	Vercel + Firebase
No AI	No model calls, no generated content

5.3 Lesson structure (locked)

Every lesson has two phases:

Scaffolded (hand-holding): 3–5 questions · hints available · explanation page after wrong answers · visual always on screen · ≥ 1 interactive question (drag/tap) · mostly multiple choice.

Mastery check (no hand-holding): 3 questions · no hints · no explanation pages until after submit · pass at $\geq 2/3$ · on fail, retry missed questions only.

5.4 Course map

#	Lesson ID	Title	x introduced?
1	lesson-01	Finding Unknowns	No (shapes only)
2	lesson-02	Variables	Yes
3	lesson-03	Finding Unknown Values	Yes
4	lesson-04	Combining and Coefficients	Yes
5	lesson-05	Writing Expressions	Yes
6	lesson-06	Two Unknowns *(optional)*	Yes

Linear unlock: complete Lesson *N* to open Lesson *N+1*. Ship Lessons 1–5 for Wednesday gate; add Lesson 6 if ahead of schedule.

5.5 Habit loop (required)

Feature	Purpose for Maya
Daily streak	Low-pressure "I practiced today" motivation
Welcome back / Continue	Reduces re-entry friction
Lesson complete screen	Satisfying finish; shows next lesson title
Course path	Clear progress without overwhelming dashboard

Not included: leagues, XP, levels, badges, push notifications, leaderboards.

5.6 Definition of done

1. Create account on phone; complete Lesson 1 with wrong answers and recovery; leave on step 3; return and resume.
2. Repeat same flow on desktop at a different viewport width.
3. Pass mastery 2/3; unlock Lesson 2; streak increments.
4. Confirm no AI endpoints in codebase or network calls.

6. User stories

6.1 In scope

ID	User story	Acceptance
US-01	Sign up with email or Google	Auth works on mobile and desktop
US-02	See display name on home	Name shown after sign-in
US-03	See course path (locked / in progress / done)	Linear list; no clutter
US-04	Tap Continue to resume exact step	Phase + step index restored
US-05	Scaffolded hints when stuck	Hint button in scaffolded only
US-06	Explanation when scaffolded answer wrong	2–4 sentences + visual; warm tone
US-07	Drag/tap scale or tiles (≥ 1 per lesson)	Touch and mouse both work
US-08	Visual updates immediately on interaction	Scale rebalances in real time
US-09	Instant feedback on submit	UI updates in $<100\text{ms}$
US-10	3-question mastery check, no hints	Pass at 2/3
US-11	Retry only missed mastery questions on fail	No full mastery restart required
US-12	Unlock next lesson on mastery pass	Next lesson becomes tappable
US-13	See daily streak on home	Updates on lesson activity
US-14	Lesson-complete celebration + next step	"Up next: ..." shown
US-15	Use app on phone in portrait	No horizontal scroll; 48px tap targets
US-16	Use app on laptop/desktop	Layout expands; mouse + keyboard work
US-17	Add lessons via JSON without rewriting UI	Content-driven renderer

6.2 Out of scope — and why

Story	Reason
AI tutor / generated problems	Phase 2; banned in MVP
Spaced repetition / review sets	Phase 3
Leagues, XP, levels	Streak + completion sufficient for Maya
Skip ahead to any lesson	Linear unlock protects scaffolding
Multiple subjects	Assignment: depth in one subject
Video lessons	Opposite of product thesis
Offline mode	Not required; adds sync complexity
Teacher dashboard	Wrong persona
Premium / payments	Not required
Push notifications	In-app streak enough for MVP
Profile photo, settings page	Maya does not need them

7. Functional requirements

7.1 Authentication

Firebase Auth: email/password and Google. Required **display name** on sign-up. Session persists across restarts. Sign out from home (no settings page).

7.2 Home screen

Display name, current streak, course path (5–6 lessons). Each card: title + status (locked / in progress / completed). Primary CTA: **Continue** or **Start**. No global nav beyond home + sign out.

7.3 Lesson player

Load lesson JSON by ID. Render step from `currentPhase` + `currentStepIndex` in Firestore. Step types: `mc`, `scale_interactive`, `tile_combine`, `explanation`. Progress indicator (e.g. "Question 2 of 4"). Scaffolded: hint visible; explanation interstitial on wrong. Mastery: no hints; brief feedback; summary at end. Auto-save on every step completion.

7.4 Feedback engine

All feedback in lesson JSON (pre-written). Client-side validation. On submit: immediate UI change then feedback copy. Scaffolded wrong → explanation interstitial before advancing. Mastery wrong → brief feedback; summarize at end.

7.5 Progress & unlock

Statuses: `locked` | `not_started` | `in_progress` | `completed`. Lesson 1 unlocked by default. Lesson *N+1* unlocks when Lesson *N* `masteryPassed === true`. Resume restores `lessonId`, `currentPhase`, `currentStepIndex`.

7.6 Streak logic

Practice day = any step completed or lesson finished. Update `lastActiveDate` (YYYY-MM-DD, local timezone). Same day: no change. Yesterday: `currentStreak += 1`. Gap ≥ 2 days: reset to 1. First activity: streak = 1. Streak 0 copy: "Start your streak today!"

7.7 Mastery pass/fail

3 questions; pass at **2/3**. On fail: encouraging copy; retry **only missed questions**. Unlimited mastery retries for MVP. On pass: mark complete, unlock next, lesson-complete screen, update streak.

8. Edge cases & error handling

8.1 Auth

Case	Behavior
Email already exists	"An account with this email already exists. Try signing in."
Wrong password	"That password doesn't match. Try again."
Session expires mid-lesson	Redirect to sign-in; resume from Firestore after re-auth
Tab closed mid-step	Last completed step saved; resume accordingly
Empty display name	Block until entered (1–30 chars)
Google sign-in, no name	Prompt for display name

8.2 Progress & sync

Case	Behavior
Same account, two devices	Firestore source of truth; latest <code>updatedAt</code> wins
Offline	Banner: "You're offline. Progress saves when you're back." Retry writes; block new lesson if no cached content
Firestore write fails	Retry once; keep local state until saved
Lesson JSON removed	Friendly error; redirect home
Complete on device A, open B	Home shows completed; no duplicate

8.3 Lesson player

Case	Behavior
Submit with no selection	Submit disabled
Double-tap submit	Debounce; ignore second
Wrong in scaffolded	Explanation interstitial; must dismiss
Wrong in mastery	Brief feedback; advance to next Q
Mastery 1/3	Retry 2 missed questions
Failed retry	Unlimited retries (MVP)
Leave mid-mastery	Save phase + step; resume same question
Incomplete drag interaction	Submit disabled until valid state
Browser back	Confirm leave or auto-save + home

8.4 Streak & responsive

Streak: first day = 1; multiple lessons/day = one increment; broken streak → "Welcome back! Let's start a new streak." Timezone: use browser local date (acceptable for MVP).

Responsive: 320px phones scale visual down without overflow; 1920px+ desktop centers content; rotate mid-lesson preserves step state; hybrid touch/mouse via pointer events; iOS Safari uses `dvh` /safe-area so CTA not hidden.

8.5 Content

Invalid JSON fails build. Missing feedback copy → fallback: "Not quite. Read the explanation and try the next one."

9. Responsive design & devices

One **responsive web app** (not native). Single codebase for all devices.

Device	Viewport	Input
Phone	320–480px portrait	Touch
Tablet	481–1024px	Touch
Desktop	1025px+	Mouse + keyboard

Element	Mobile	Desktop
Course path	Single column, full width	Centered, max ~720px
Lesson player	Visual above prompt	Stacked or side-by-side, max ~800px
Scale/tile visual	Full width, min 200px	Larger, max ~500px centered
MC options	Stacked buttons, min 48px	Stacked or 2-column
Typography	16px base (prevents iOS zoom)	16–18px base

Interaction parity: scale drag via pointer events (touch + mouse); no hover-only affordances; Tab + Enter for MC on desktop; no horizontal scroll; resize/rotate preserves lesson state.

Maya does not need: native app install, separate mobile/desktop codebases, desktop-only shortcuts beyond accessibility.

10. Non-functional requirements

Requirement	Target
Feedback UI latency	<100ms from submit to visual change
Lesson load	<2s to first interaction on 4G
Animation	60fps goal on scale/tile updates
Concurrent users	Firebase default scaling
Security	Firestore: users read/write own data only
Accessibility	Semantic HTML, focus states, contrast, keyboard MC
AI	Zero model API calls in Phase 1
Browsers	Latest Chrome, Safari, Firefox, Edge

11. Tech stack

Required

Layer	Technology
Frontend	React (Vite)
Auth	Firestore Authentication
Database	Cloud Firestore
Hosting	Vercel

Recommended

Layer	Technology
Styling	Tailwind CSS
Routing	React Router
State	React Context + hooks
Visuals	SVG + React
Content	JSON in /content

Not in stack (Phase 1)

OpenAI / Anthropic / any LLM · Supabase · custom Node server · Canvas/WebGL · native wrappers (Capacitor, React Native).

12. Data schema

12.1 Storage overview

Data	Where	Mutable?
Lessons, feedback, explanations	JSON in /content	Deploy to update
User identity	Firestore Auth	Yes
Profile, streak	Firestore users/{uid}	Yes
Lesson progress	Firestore users/{uid}/progress/{lessonId}	Yes

12.2 users/{userId}

```
displayName: string      // required
email: string
createdAt: Timestamp
lastActiveDate: string  // "2026-06-22" local date
currentStreak: number   // default 0
updatedAt: Timestamp
```

12.3 users/{userId}/progress/{lessonId}

Document ID = lessonId (e.g. lesson-01).

```
lessonId: string
status: "not_started" | "in_progress" | "completed"
currentPhase: "scaffolded" | "mastery"
currentStepIndex: number // 0-based within phase
scaffoldedCompleted: boolean
masteryCorrectCount: number
masteryMissedStepIds: string[]
masteryPassed: boolean
startedAt: Timestamp | null
completedAt: Timestamp | null
updatedAt: Timestamp
```

12.4 Firestore security rules

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /users/{userId} {
      allow read, write: if request.auth != null && request.auth.uid == userId;
      match /progress/{lessonId} {
        allow read, write: if request.auth != null && request.auth.uid == userId;
      }
    }
  }
}
```

12.5 Not stored

Individual step attempt history · AI conversation logs · analytics events (skip for MVP unless needed for demo).

13. Information architecture

Route	Screen
/auth	Sign in / sign up + display name
/	Home: name, streak, path, Continue
/lesson/:lessonId	Lesson player
(overlay)	Lesson complete: celebration + "Up next"

4 screens total. No settings, profile edit, or about page.

14. Lesson content model

File structure

```
content/
  course.json
  lessons/
    lesson-01.json ... lesson-05.json
```

course.json

```
{
  "id": "solving-equations-intro",
  "title": "Solving Equations",
  "lessonOrder": ["lesson-01", "lesson-02", "lesson-03", "lesson-04", "lesson-05"]
}
```

Step schema (per question)

Field	Purpose			
id	Unique step ID			
type	mc \	scale_interactive \	tile_combine \	explanation
prompt	Question text			
visual	Scale/tile config object			
options / correctIndex	MC answers			
validation	Rules for interactive steps			
hint	Scaffolded only			
feedback	{ correct, incorrect[] }			
explanation	{ title, body, visual } — scaffolded wrong answers			

Example step (abbreviated)

```
{
  "id": "s2",
  "type": "scale_interactive",
  "prompt": "Remove 1 block from each side to find the unknown.",
  "visual": { "type": "scale", "config": { "left": ["unknown", "1"], "right": ["7"] } },
  "validation": { "action": "remove_from_both", "value": 1, "expectedUnknown": 6 },
  "hint": "Whatever you remove from one side, remove from the other too.",
  "feedback": {
    "correct": "The scale stays balanced!",
    "incorrect": ["You changed only one side – both sides must stay equal."]
  }
}
```

15. Course outline (per lesson)

Lesson 1 — Finding Unknowns: Equal means equal; shapes on scales. No x. Scaffolded: 4 Q (2 MC, 1 scale interactive, 1 MC). Mastery: 3 MC.

Lesson 2 — Variables: x names an unknown. Scaffolded: 4 Q (introduce x , evaluate $x + 3$, interactive label match, MC). Mastery: 3 Q.

Lesson 3 — Finding Unknown Values: One-step equations via balance reasoning. Scaffolded: 4 Q ($x + 3 = 7$ interactive, $x - 2 = 6$, $10 = x + 4$, mistake explanation). Mastery: 3 mixed $+/ -$.

Lesson 4 — Combining and Coefficients: $2x + x = 3x$; combine then solve; coeffs ≤ 3 . Scaffolded: 4 Q. Mastery: 3 combine-and-solve.

Lesson 5 — Writing Expressions: Words $\rightarrow$ algebra; watch "less than" order trap. Scaffolded: 4 Q. Mastery: 3 word $\rightarrow$ expression MC.

Lesson 6 (optional) — Two Unknowns: Substitute when one known; no coefficients. Scaffolded: 3–4 Q. Mastery: 3 Q.

16. Out of scope

Phase 1: AI · leagues/XP/levels · push notifications · offline · educator tools · premium · multiple courses · profile editing · step analytics.

Phase 2 (later): AI hints, problem generation, adaptive path — document in Brainlift.

Phase 3 (later): Spaced repetition, interleaving, advanced mastery — document in Brainlift.

17. Success criteria & test scenarios

#	Scenario	Pass
1	Lesson 1 E2E with wrong answers + recovery	Hints/explanations help Maya finish
2	Manipulate interactive; visual responds live	Scale/tiles update on drag/tap
3	Leave mid-lesson; return	Same step; streak persists
4	Finish lesson	"Up next: Variables" (or next title)
5	Phone 375px	No broken layout
6	Desktop 1280px	Mouse interactions work

Additional acceptance: sign up $\rightarrow$ streak=1 · mastery 2/3 unlocks next · cross-device sync · offline banner · no AI in DevTools · all 5 lessons playable.

18. Open questions & build order

Question	Recommendation
Product name	Unknown (decided)
Vite vs Next.js	Vite (decided)
Lesson 6 in MVP?	Ship 5; add 6 if ahead (decided)
Email + Google?	Both (decided)

Build order: scaffold $\rightarrow$ Lesson 1 JSON + scale component $\rightarrow$ renderer + feedback $\rightarrow$ auth $\rightarrow$ Firestore progress $\rightarrow$ home/path/streak $\rightarrow$ Lessons 2–5 $\rightarrow$ edge cases + responsive polish $\rightarrow$ deploy.

README must state: **Unknown** · Algebra · Maya persona · setup guide · architecture (React + Firebase + JSON) · deployed link.

Phase 1 only. Clone Brilliant's core learn-by-doing mechanics; skip everything Maya doesn't need. If a feature does not help her learn algebra, persist progress, or come back tomorrow — it does not ship.